

CS 471 – Lesson 8 Solutions Key

Part 1: Adversarial Search (Minimax) Solutions

1. Layer 4 & 3 Evaluation:

Solution

Answer:

- **Min nodes (L4):** $G=3, H=5, I=2, J=1, K=6, L=2, M=4, N=3$.
- **Max nodes (L3):** $C=\max(3, 5) = 5; D=\max(2, 1) = 2; E=\max(6, 2) = 6; F=\max(4, 3) = 4$.

2. Layer 2 Evaluation:

Solution

Answer:

- Min Agent A minimizes between $C(5)$ and $D(2)$, choosing **2**.
- Min Agent B minimizes between $E(6)$ and $F(4)$, choosing **4**.

3. Root Node & Policy:

Solution

Answer: The root node (S) maximizes between $A(2)$ and $B(4)$. The final value is **4**. The Max agent should take the path towards **Node B**.

4. Theory:

Solution

Answer: Minimax models the **worst-case perspective** because it assumes the opponent is a perfect player making optimal moves to minimize your score[cite: 3].

Part 2: Stochastic Search (Expectimax) Solutions

5. Layer 4 Expected Values:

Solution

Answer: Values reflect average-case outcomes, calculated by $v = p_1(\text{val}_1) + p_2(\text{val}_2)$ [cite: 3].

- **C1:** $0.2(8) + 0.8(4) = 1.6 + 3.2 = 4.8$
- **C2:** $0.4(10) + 0.6(12) = 4.0 + 7.2 = 11.2$
- **C3:** $0.7(0) + 0.3(6) = 0.0 + 1.8 = 1.8$
- **C4:** $0.9(14) + 0.1(2) = 12.6 + 0.2 = 12.8$
- **C5:** $0.5(7) + 0.5(3) = 3.5 + 1.5 = 5.0$
- **C6:** $0.1(9) + 0.9(15) = 0.9 + 13.5 = 14.4$
- **C7:** $0.3(20) + 0.7(10) = 6.0 + 7.0 = 13.0$
- **C8:** $0.6(2) + 0.4(4) = 1.2 + 1.6 = 2.8$

6. Layer 3 Max Values:

Solution

Answer: Max nodes choose the highest available expected utility from their children[cite: 3].

- **M1:** $\max(4.8, 11.2) = 11.2$
- **M2:** $\max(1.8, 12.8) = 12.8$
- **M3:** $\max(5.0, 14.4) = 14.4$
- **M4:** $\max(13.0, 2.8) = 13.0$

7. Layer 2 Expected Values:

Solution

Answer:

- **X:** $0.4(11.2) + 0.6(12.8) = 4.48 + 7.68 = 12.16$
- **Y:** $0.8(14.4) + 0.2(13.0) = 11.52 + 2.60 = 14.12$

8. Root Node (Max):

Solution

Answer: The final Expectimax value at the Root is $\max(12.16, 14.12) = 14.12$. The optimal policy for the blue-force drone is to take the path towards **Node Y**, as it yields the highest average-case outcome under optimal play[cite: 3].